Stellenbosch
UNIVERSITY
IYUNIVESITHI
UNIVERSITEIT

Machine Learning 441/741

Assignment 1: Data Exploration

Total: [150]
Deadline: 10 August 2026, 08:00

Instructions

When completing this assignment, follow the instructions given below:

- The assignment must be completed by each student individually.
- You may use any programming language and software tools to complete the assignment. You may make use of any libraries.
- You will submit this assignment on STEMlearn by uploading tabular data, which will provide your responses to all of the tasks listed in the assignment. Note that while some coding will be required, your code will not be evaluated.
- The data that is uploaded at the time of the deadline will be accepted as your final submission.
- No late submissions will be accepted.
- Generative AI tools may not be used.

Purpose of the Assignment

The main purpose of this assignment is to evaluate your skills at analyzing a given data set, with the main goals to characterize the data set, to conduct an exploratory analysis of the dataset, and to use the outcome of such analysis to identify data quality issues within the dataset.

For the purpose of this assignment, a dataset is available from the Assignment Specifications section on STEMlearn. Download the data set, `badDataSet.zip`, uncompress the `.zip` file and start with your exploration of the `badDataSet.csv` file.

Note the following about the dataset:

- The file is tab delimited.
- The real descriptive feature names have been removed to ensure that you have a complete unbiased analysis of the dataset. Your very first step is to label the descriptive features from the first column to the second last column as A_i, where i refers to the column number. The last column refers to the target feature, which you will simply label as T.
- Missing values are indicated using the character '?'.
- This is a classification problem.
- A cleaned version of this dataset will be used for the next assignments, so it is important that you gain a good understanding of the dataset.

On the uploading of tabular data

All tabular data should be uploaded in comma delimited `.csv` format. In all cases the column order is given and must be strictly adhered to. Any deviation from the given column order may result in a total loss of marks. When saving the tabular data to file, do not save the index nor the column names to the file. After saving tabular data to file, ensure that the command `pd.read_csv(<file name>, header=None, names=<given columns>)` results in the expected output.

Task 1:Data Quality Report

Your first task is to construct the data quality report (DQR) for this dataset.

1. Upload the DQR tables for all of the descriptive features.
   - Upload the DQR tables for the continuous and the categorical features separately.
   - Column order:
     - Continuous: Feature, Count, % Miss., Card., Min., 1st Qrt., Mean, Median, 3rd Qrt., Max., Std. Dev.
     - Categorical: Feature, Count, % Miss., Card., Mode, Mode Freq., Mode %, 2nd Mode, 2nd Mode Freq., 2nd Mode %
   - Note that marking will be done by subtracting 0.5 marks from the total for this question per missing item or error in the DQR.
2. Upload histograms for all the continuous descriptive features, and bar plots for all the categorical descriptive features.
   - Upload one `.pdf` file for all the histograms, and one `.pdf` file for all the bar plots.
   - Each plot should be as small as possible, while still being legible and with all axes labels visible. Organise the plots into a matrix of some number of rows and columns.
   - Note that marking will be done by subtracting 0.5 marks from the total for this question per missing figure or incorrect type of figure.

(22)

3. Upload the DQR table for the target feature.
   - Use the column order as given in the previous question.
4. Upload either a histogram or bar plot for the target feature.
   - The type of plot depends on the type of the target feature.
   - The filetype should be `.pdf`.

(2)

Task 2: Data Quality Issues & Handling Strategies

The second task requires that you identify data quality issues within the dataset, and provide strategies to handle them. Note that you will only identify and discuss the data quality issues, and that you are not required to correct the data quality issues. A cleaned dataset will be provided for the purposes of the next assignments.

When suggesting handling strategies, do not refer to specific machine learnings algorithms.

For this a table as illustrated below should be created.

Feature	Data Quality Issue	Evidence	Handling Strategy	Justification
A1	Name the issue	Evidence of issue	Solution for issue	Justification of solution
A2	Name the issue	Evidence of issue	Solution for issue	Justification of solution
...	...	...	...	...

1. Upload the completed table with all of the data quality issues and handling strategies that you have identified.
    - The values in the 'Feature' column should be the name of a single column - create multiple rows for features with the same data quality issues.
    - The values in column 'Data Quality Issue' **must be** selected from those in `dataQualityIssue.csv`, which is uploaded on STEMlearn.
    - Your 'Evidence' column must provide evidence-based reasons for why you believe that the data quality issue is present. Evidence can be references to information in the DQR(s), or from tests, distribution plots, correlation plots, or any other form of data visualization.
    - The values in column 'Handling Strategy' **must be** selected from those in `handlingStrategy.csv`, which is uploaded on STEMlearn.
    - The 'Justification' column must provide an adequate justification for the handling strategy that has been proposed.
    - Note that the evidence and justification columns are the most important part of this task, and weights to most with reference to the mark allocation.
    - A feature may exhibit more that one data quality issue.
    - A data quality issue may have more than one viable handling strategy.
    - For the purposes of marking, assume that stating $A \leftrightarrow B$ is equivalent to stating $B \leftrightarrow A$.

(126)

Mark Rubric

The assignment will be assessed as follows:

Aspect		Mark
Task 1		
	Descriptive features data quality report table and figures	22
	Target feature data quality report and figure	2
Task 2		
	Data quality issues	24
	Evidence	48
	Handling Strategy	18
	Justification	36
	Total	150